Asymmetric Cryptography

Applied Field Theory 101

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Introduction

1.1 Terminology

Include public-key encryption and signature table with direction.

Alice, Bob, Mallory.

1.2 Trapdoors

1.3 Field Theory

Rivest–Shamir–Adleman

2.1 Description

Key Generation:

1. Generate two large distinct prime numbers p and q.
2. Calculate $\lambda(pq) = \text{lcm}(p-1, q-1)$.
3. Calculate e such that e and $\lambda(pq)$ are coprime and $2 < e < \lambda(pq)$.
4. Calculate d such that $de \mod \lambda(pq) = 1$.

The public key is (pq, e) and the private key is d, $\lambda(pq)$ can be discarded.

Analysis:

There's a multiple applications for the RSA cryptosystem so instead of jumping into any particular one we'll focus on the key feature, that for an arbitrary $0 \leq m < pq$ we have:

$$m^{ed} \mod pq = m$$

This is because by construction $ed - 1$ is a multiple of $p - 1$ meaning:

$$m^{ed} \mod p = m^{ed-1}m \mod p = m \mod p$$

Remark:

Observe that the above proof doesn't really use any of $\lambda(pq)$'s properties beyond it being divisible by $p - 1$ and $q - 1$. When $m \in (\mathbb{Z}/pq\mathbb{Z})^\times$ we can present a much cooler proof. In this case $ed - 1$ is a multiple of $\lambda(pq)$, hence:

$$m^{ed-1} \mod pq = 1$$

From which the main result follows.

The above proof is often presented as the sole proof of correctness for RSA, this is incorrect and probably done because it is much cooler.

In practice it is true most of the time since p and q are large and:

$$\frac{|(\mathbb{Z}/pq\mathbb{Z})^\times|}{|\mathbb{Z}/pq\mathbb{Z}|} = \frac{\phi(pq)}{pq} = 1 - \frac{1}{p} - \frac{1}{q} + \frac{1}{pq}$$

2.2 Applications

Elliptic Curve Cryptography

3.1 Elliptic Curves

3.2 Elliptic Curve Digital Signature Algorithm

3.3 Remarks

playstation and Android wallet

Diffie–Hellman Key Exchange

The Diffie-Hellman Key Exchange is a short but sweet demonstration of asymmetric cryptography. The goal is for Alice and Bob to use a public channel to end with a shared private key.

1. Alice and Bob agree on an integer $n > 1$ and a generating element g of the cyclic group C_n in private.
2. Independently and in private, Alice and Bob pick random integers $1 < a < n$ and $1 < b < n$ respectfully.
3. Alice and Bob let g^a and g^b be publicly known.
4. Alice and Bob calculate $(g^b)^a$ and $(g^a)^b$ from the public g^b and g^a respectfully.

Since:

$$(g^b)^a = g^{ba} = g^{ab} = (g^a)^b$$

Alice and Bob agree on g^{ab} while only making n, g, g^a and g^b public. Under the discrete logarithm trapdoor assumption, a third party can not deduce a or b from this information meaning g^{ab} is safe from attackers.

Two important remarks.

Firstly, if g is publicly known then the exchange is susceptible to a man-in-the-middle attack. Mallory is able to pretend to be Alice to Bob and send g^m then Mallory and Bob will share the key g^{mb} with Bob thinking the key was shared with Alice.

Secondly, The most direct implementation uses the $G = (\mathbb{Z}/p\mathbb{Z})^\times$ with prime p and is called the Finite Field Diffie-Hellman. This is because $\mathbb{Z}/p\mathbb{Z}$ is a finite field and $(\mathbb{Z}/p\mathbb{Z})^\times$ is always guaranteed to be a cyclic group.

Merkle–Hellman Knapsack

The knapsack problem is an easy to grasp yet complex to solve optimization problem best summarised through the archetypal example:

> Given a knapsack with with a limited size and a collection of items of fixed size and value determine which selections of items with a combined size less than or equal to the knapsack has the maximum value.

The practical applications are obvious and the problem has been thoroughly studied because of it. Yet the problem remains difficult to compute for arbitrary items, which makes the knapsack problem a candidate for producing a trapdoor function.

For a trapdoor function to work we need a version of the problem which is hard without private information but easy with private information. The Subset Sum Problem (SSP) is a special case of the knapsack problem where the items values are constant and we must find a collection that **equals** the knapsack size, not just be less than. With arbitrary item sizes the SSP is known to be hard, but when the item sizes are superincreasing the problem is trivial.

A sequence $\{w_k\}$ is superincreasing if we have:

$$w_n > \sum_{k=1}^{n-1} w_k$$

In our terms it means there is no combination of items with size less than a given item that have a combined size greater or equal to the first item. This makes solving the superincreasing SSP easy because if we always take the

largest item that would fit in the remaining knapsack size, since the remaining smaller items can't make up the difference.

Hence the plan is to construct some private key which can convert a public SSP to a private superincreasing SSP.

5.1 Description

Key Generation:

Let n be the desired block size.

1. Generate a random superincreasing n-array $\{w_k\}$.
2. Generate a random integer q such that:

$$q > \sum_{k=1}^{n} w_k$$

3. Generate a random integer r that is coprime to q.
4. Calculate a new n-array $\{b_k\}$ such that:

$$b_k = rw_k \mod q$$

5. Calculate r' such that:

$$rr' \mod q = 1$$

The public key is $\{b_k\}$ and the private key is $(\{w_k\}, q, r')$, observe that r can be discarded.

Encryption:

Let the n-bit message be given by the bits m_k. The cipher text is given by:

$$c = \sum_{k=1}^{n} m_k b_k$$

Decryption:

Treat $(r'c \mod q)$ and $\{w_k\}$ as the knapsack size and item sizes for a super-increasing SSP problem, w_k will be in the knapsack if and only if $m_k = 1$.

5.2 Remarks

Appendix

A.1 Algebra Revision

A.1.1 Lagrange's Theorem

A.1.2 Cyclic Group

A.1.3 Bézout's Identity

I've send multiple time "Calculate x such that $yx \mod z = 1$

A.1.4 Chinese Remainder Theorem

The Chinese Remainder Theorem is an interesting bring between regular and abstract algebra and I'll be using both forms in this document.

Regular:

Let n_1 and n_2 be coprime integers, from Bézout's identity there exists q_1 and q_2 such that:

$$q_1 n_1 + q_2 n_2 \mod n_1 n_2 = 1$$

Now consider another integer x such that:

$$x \mod n_1 = a_1 \text{ and } x \mod n_2 = a_2$$

Then the only $0 \leq x < n_1 n_2$ is given by

$$x = a_1 n_2 q_2 + a_2 n_1 q_1 \mod n_1 n_2$$

Proof follows by substitution.

Consider the corollary when $a = a_1 = a_2$ then we have:

$$\begin{aligned} x =& a_1 n_2 q_2 + a_2 n_1 q_1 \mod n_1 n_2 \\ =& a(n_2 q_2 + n_1 q_1) \mod n_1 n_2 \\ =& a \end{aligned}$$

Abstract:

We can use the prior construction to define a function

$$\begin{aligned} f : \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} &\to \mathbb{Z}/n_1 n_2\mathbb{Z} \\ (a_1, a_2) &\mapsto a_1 n_2 q_2 + a_2 n_1 q_1 \end{aligned}$$

It's left as an excise that f is an isomorphism, meaning:

$$\mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \cong \mathbb{Z}/n_1 n_2\mathbb{Z}$$

You can use induction to generalize this to:

$$\mathbb{Z}/N\mathbb{Z} \cong \prod_k \mathbb{Z}/n_k\mathbb{Z}$$

Where n_k are pairwise relatively prime positive integers and:

$$N = \prod_k n_k$$

Of particular importance is the case where $n_k = p_k^{m_k}$ for distinct primes p_k and integers m_k.

A.1.5 Multiplicative group of integers modulo n

The "multiplicative group of integer module n" is is the multiplicative group of the ring $\mathbb{Z}/n\mathbb{Z}$ and is denoted $(\mathbb{Z}/n\mathbb{Z})^\times$.

Structure:

For general rings R and S we have:

$$(R \times S)^\times \cong R^\times \times S^\times$$

This follows from the, some what awkward looking, function:

$$f : (R \times S)^\times \to R^\times \times S^\times$$
$$(r, s) \mapsto (r, s)$$

Applying this to the Chinese remainder theorem gives:

$$(\mathbb{Z}/n\mathbb{Z})^\times \cong \prod_k (\mathbb{Z}/p_k^{m_k}\mathbb{Z})^\times$$

Where n is given by the distinct primes p_k as:

$$n = \prod_k p_k^{m_k}$$

When $p \neq 2$ we have:

$$(\mathbb{Z}/p^m\mathbb{Z})^\times = C_{p^m - p^{m-1}}$$

And when $p = 2$ we have:

$$(\mathbb{Z}/2^m\mathbb{Z})^\times = \begin{cases} C_2 \times C_{2^{m-2}} & m \geq 2 \\ C_1 & m = 1 \end{cases}$$

The general m case is outside the scope of this document, but can be found in most undergrad algebra textbooks. The $m = 1$ case is pretty tractable:

Sketch of the $m = 1$ case:

The goal is to prove that:

$$(\mathbb{Z}/p\mathbb{Z})^\times$$

Is cyclic, the structure of this proof is the following:

1. Prove there is some $d \in (\mathbb{Z}/p\mathbb{Z})^\times$ such that $g \in (\mathbb{Z}/p\mathbb{Z})^\times \Rightarrow g^{\mathrm{ord}(d)} = 1$.
2. Hence, the polynomial $f(x) = x^{\mathrm{ord}(d)} - 1$ has $|(\mathbb{Z}/p\mathbb{Z})^\times)|$ distinct roots.
3. Hence, $\mathrm{ord}(d) = |(\mathbb{Z}/p\mathbb{Z})^\times|$.[1]
4. Hence, $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic with generator g.

[1] This only works because $\mathbb{Z}/p\mathbb{Z}$ is a field.

It takes a few steps to prove such a d exists, an important result to help us is that:

$$\operatorname{ord}(g^n) = \frac{\operatorname{ord}(g)}{\gcd(\operatorname{ord}(g), n)}$$

This result is pretty common in algebra textbooks, but a sketch of the proof is that by the minimality of $\operatorname{ord}(g)$ and $\operatorname{ord}(g^n)$ the relation:

$$(g^n)^{\operatorname{ord}(g^n)} = g^{n \operatorname{ord}(g^n)} = 1$$

Means every factor of $\operatorname{ord}(g)$ not in n has to come from $\operatorname{ord}(g^n)$.
There is three corollaries from this result:

Inverse Corollary:

From the $n = -1$ case we have:

$$\operatorname{ord}(g^{-1}) = \operatorname{ord}(g)$$

Relatively Prime Corollary:

Let $x, y \in (\mathbb{Z}/p\mathbb{Z})^\times$ with $\operatorname{ord}(x)$ and $\operatorname{ord}(y)$ being relatively prime, then:

$$\operatorname{ord}(xy) = \operatorname{ord}(x) \operatorname{ord}(y)$$

Let $(xy)^n = 1$, since $(\mathbb{Z}/p\mathbb{Z})^\times$ is Abelian we have:

$$x^n y^n = (xy)^n = 1$$

From the previous corollary we have:

$$\operatorname{ord}(x^n) = \operatorname{ord}(y^n)$$

Giving:

$$\frac{\operatorname{ord}(x)}{\gcd(\operatorname{ord}(x), n)} = \frac{\operatorname{ord}(y)}{\gcd(\operatorname{ord}(y), n)}$$

The LHS is a factor of $\operatorname{ord}(x)$ and likewise for the RHS and $\operatorname{ord}(y)$. Since they are relatively prime they can be equal only when they are both 1, meaning:

$$\operatorname{ord}(x) = \gcd(\operatorname{ord}(x), n), \quad \operatorname{ord}(y) = \gcd(\operatorname{ord}(y), n)$$

Hence n is a multiple of both $\operatorname{ord}(x)$ and $\operatorname{ord}(y)$. Since $\operatorname{ord}(x)$ and $\operatorname{ord}(y)$ are relatively prime the smallest such n is:

$$n = \operatorname{ord}(x) \operatorname{ord}(y) \quad \square$$

Lowest Common Denominator Corollary:

Now let $x, y \in (\mathbb{Z}/p\mathbb{Z})^\times$ and let:

$$g = \gcd(\operatorname{ord}(x), \operatorname{ord}(y))$$

Then:

$$\operatorname{ord}(x^g y) = \operatorname{lcm}(\operatorname{ord}(x), \operatorname{ord}(y))$$

This follows from $\operatorname{ord}(x^g)$ and $\operatorname{ord}(y)$ being relatively prime as:

$$\begin{aligned}\operatorname{ord}(x^g) =& \frac{\operatorname{ord}(x)}{\gcd(\operatorname{ord}(x), g)} \\ =& \frac{\operatorname{ord}(x)}{g}\end{aligned}$$

Means $\operatorname{ord}(x^g)$ is $\operatorname{ord}(x)$ with all its common factors with $\operatorname{ord}(y)$ removed.

Hence we can use the previous corollary to get:

$$\begin{aligned}\operatorname{ord}(x^g y) =& \operatorname{ord}(x^g)\operatorname{ord}(y) \\ =& \frac{\operatorname{ord}(x)\operatorname{ord}(y)}{g} \\ =& \operatorname{lcm}(\operatorname{ord}(x), \operatorname{ord}(y)) \quad \square\end{aligned}$$

Proving d exists:

From the preceding corollary for any $x, y \in (\mathbb{Z}/p\mathbb{Z})^\times$ there exits $z \in (\mathbb{Z}/p\mathbb{Z})^\times$ such that:

$$\operatorname{ord}(z) = \operatorname{lcm}(\operatorname{ord}(x), \operatorname{ord}(y))$$

Now $\operatorname{lcm}(\operatorname{ord}(x), \operatorname{ord}(y))$ is by definition a multiple of $\operatorname{ord}(x)$ and $\operatorname{ord}(y)$ hence:

$$z^{\operatorname{ord}(z)} = x^{\operatorname{ord}(z)} = y^{\operatorname{ord}(z)} = 1$$

Hence a there exists $d \in (\mathbb{Z}/p\mathbb{Z})^\times$ such that:

$$g \in (\mathbb{Z}/p\mathbb{Z})^\times \Rightarrow g^{\operatorname{ord}(d)} = 1$$

Since we can take two elements x and y and calculate $z = x^{\gcd(\operatorname{ord}(x),\operatorname{ord}(y))}y$, then we take another element w and calculate $z^{\gcd(\operatorname{ord}(z),\operatorname{ord}(x))}x^{\gcd(\operatorname{ord}(x),\operatorname{ord}(y))}y$. We continue until all elements of $(\mathbb{Z}/p\mathbb{Z})^\times$ are included, the end result of this process is d.

Cyclic Corollary:

A corollary of the structure of $(\mathbb{Z}/n\mathbb{Z})^\times$ is that it cyclic if and only if:

$$n = 1, 2, 4, p^k, 2p^k$$

For some odd prime p.

This is because the product of cyclic groups is cyclic if and only if all but one are of size 1.

Example $(\mathbb{Z}/9\mathbb{Z})^\times$ table:

Let $g \in (\mathbb{Z}/9\mathbb{Z})^\times$, bellow is a table of important properties that is useful as an example:

g	g^2	g^3	g^4	g^5	g^6	$\text{ord}(g)$	g^{-1}
1	1	1	1	1	1	1	1
2	4	8	7	5	1	6	5
4	7	1	4	7	1	3	7
5	7	8	4	2	1	6	2
7	4	1	7	4	1	3	4
8	1	8	1	8	1	2	8

A.1.6 Totient Functions

A totient of an integer n is a number less than n that is relatively prime to n. Totient are important to cryptography because they are, by definition, the elements of $(\mathbb{Z}/n\mathbb{Z})^\times$.

From this group we define two functions, Euler's totient function:

$$\phi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|$$

and the Carmichael function[2]:

$$\lambda(n) = \text{lcm}\left\{ \text{ord}(k) \mid k \in (\mathbb{Z}/n\mathbb{Z})^\times \right\}$$

These functions are often studied in their own right

[2] I have not idea why Euler gets to own his function but Carmichael does not.

Euler's Product Formula:

Carmichael's Theorem:

Carmichael's Theorem:

Remark:

A.2 Subset Sum Problem

The Subset Sum Problem (SSP) is the type of problem with a lot of defined and studied variants. So it's worth formalizing our variant is:

> Given a nonincreasing positive sequence $\{w_k\}_{k=1}^n$ and an integer c there is a sequence $\{m_k\}_{k=1}^n \in \{0,1\}^n$ such that:
>
> $$c = \sum_{k=1}^{n} m_k w_k$$
>
> return such a sequence.

We will formalize the greedy SSP algorithm as the recursive function:

$$gSSP(c, \{w_k\}_{k=1}^n) = \begin{cases} \emptyset & n = 0 \\ \{1\}^\frown gSSP(c - w_k, \{w_k\}_{k=1}^{n-1}) & c \geq w_n \\ \{0\}^\frown gSSP(c, \{w_k\}_{k=1}^{n-1}) & c < w_n \end{cases}$$

This algorithm is called greedy because at each call it reduces the first argument, if it is able to. **This does not work in the general SSP case!**

For example, consider $gSSP(5, \{4,3,2\}) = \{4\}$, the algorithm greedily takes 4 but is then unable to make up the $5 - 4 = 1$ difference. But if it has waited it would have found $2 + 3 = 5$.

But I've claimed it does work then $\{w_k\}$ is superincreasing, this appendix is meant to back that up.

A.2.1 Proof of Correctness

Correctness equates to proving:

$$gSSP\left(\sum_{k=1}^{n} m_k w_k, \{w_k\}_{k=1}^n\right) = \{m_k\}_{k=1}^n$$

This will be done by induction, but first some useful lemmas.

Superincreasing lemmas:

If $\sum_{k=1}^{n} m_k w_k \geq w_n$ then $m_n = 1$.

Proof, assume $m_n = 0$ then:

$$\sum_{k=1}^{n} m_k w_k = \sum_{k=1}^{n-1} m_k w_k \leq \sum_{k=1}^{n-1} w_k < w_n$$

A contradiction.

If $\sum_{k=1}^{n} m_k w_k < w_n$ then $m_n = 0$.

Proof, assume $m_n = 1$ then:

$$\sum_{k=1}^{n} m_k w_k \geq w_n$$

A contradiction.

Base Case $n = 1$:

$$\begin{aligned}
gSSP\left(\sum_{k=1}^{1} m_k w_k, \{w_k\}_{k=1}^{1}\right) =& gSSP\left(m_1 w_1, \{w_1\}\right) \\
&= \begin{cases} \{1\}^\frown gSSP((m_1 - 1)w_1, \emptyset) & m_1 w_1 \geq w_1 \\ \{0\}^\frown gSSP(m_1 w_1, \emptyset) & m_1 w_1 < 1_1 \end{cases} \\
&= \begin{cases} \{1\}^\frown \emptyset & m_1 w_1 \geq w_1 \\ \{0\}^\frown \emptyset & m_1 w_1 < 1_1 \end{cases} \\
&= \begin{cases} \{1\} & m_1 w_1 \geq w_1 \\ \{0\} & m_1 w_1 < 1_1 \end{cases}
\end{aligned}$$

If $m_1 = 1$ we have:

$$m_1 w_1 \geq w_1 \Rightarrow gSSP(m_1 w_1, \{w_1\}) = \{1\} = \{m_1\}$$

else if $m_1 = 0$ and we have:

$$m_1 w_1 < w_1 \Rightarrow gSSP(m_1 w_1, \{w_1\}) = \{0\} = \{m_1\}$$

As required

Induction:

The n case is given by the verbose:

$$gSSP\left(\sum_{k=1}^{n} m_k w_k, \{w_k\}_{k=1}^{n}\right) = \begin{cases} \{1\}^\frown gSSP\left(\sum_{k=1}^{n} m_k w_k - w_k, \{w_k\}_{k=1}^{n-1}\right) & \sum_{k=1}^{n} m_k w_k \geq w_n \\ \{0\}^\frown gSSP\left(\sum_{k=1}^{n} m_k w_k, \{w_k\}_{k=1}^{n-1}\right) & \sum_{k=1}^{n} m_k w_k < w_n \end{cases}$$

It can be simplified using the superincreasing lemmas, if $\sum_{k=1}^{n} m_k w_k \geq w_n$ we have $m_n = 1$ and:

$$\sum_{k=1}^{n} m_k w_k - w_k = \sum_{k=1}^{n-1} m_k w_k$$

and if $\sum_{k=1}^{n} m_k w_k < w_n$ we have $m_n = 0$ and:

$$\sum_{k=1}^{n} m_k w_k = \sum_{k=1}^{n-1} m_k w_k$$

Combining these results gives:

$$gSSP\left(\sum_{k=1}^{n} m_k w_k, \{w_k\}_{k=1}^{n}\right) = \{m_n\}^\frown gSSP\left(\sum_{k=1}^{n-1} m_k w_k, \{w_k\}_{k=1}^{n-1}\right)$$

Which by the induction assumption:

$$gSSP\left(\sum_{k=1}^{n} m_k w_k, \{w_k\}_{k=1}^{n}\right) = \{m_n\}^\frown \{m_k\}_{k=1}^{n-1} = \{m_k\}_{k=1}^{n}$$

As required.

A.2.2 Proof of Uniqueness

For a given superincreasing sequence $\{w_k\}_{k=1}^{n}$ assume there are two distinct sequences $\{n_k\}_{k=1}^{n}$ and $\{m_k\}_{k=1}^{n}$ in $\{0,1\}^n$ such that:

$$\sum_{k=1}^{n} m_k w_k = \sum_{k=1}^{n} n_k w_k$$

Without loss of generality assume that:

$$n_k \neq m_k$$

Let K be the largest integer such that $m_k = 1$ or $n_k = 1$. Without loss of generality assume that it is $m_K = 1$, and have:

$$\sum_{k=1}^{K} m_k w_k = \sum_{k=1}^{K-1} n_k w_k$$

However:

$$\sum_{k=1}^{K} m_k w_k \geq w_K > \sum_{k=1}^{K-1} w_k \geq \sum_{k=1}^{K-1} n_k w_k$$

A contradiction, hence $\{n_k\}_{k=1}^{n}$ and $\{m_k\}_{k=1}^{n}$ are not distinct.

Uniqueness is an important property for our application since it ensures decryption always gives the original message.

A.2.3 Digits

Consider the formula for geometric progression:

$$\sum_{k=0}^{n} r^n = \frac{r^{n+1} - 1}{r - 1}$$

If $r > 1$ this formula proves $w_k = r^{k-1}$ is a superincreasing function, in fact we get the stronger:

$$r^{n+1} > (r-1) \sum_{k=0}^{n} r^n$$

With simple modifications the results for the superincreasing lemmas and uniqueness sections can be generalized to allow $\{m_k\}_{k=1}^{n} \in \{0, 1, 2, ..., r-1\}^n$.

It should have now hit you that under this generalized an expression $\sum_{k=1}^{n} m_k r^{k-1}$ is just the digit expansion. And further that the $gSSP$ algorithm is just reading digits left-to-right. So this whole section is proving something children implicitly understand once they start using digits, that's math for ya.

A.3 Montgomery Ladder